

HSIU-HSUAN YEH (葉秀軒) *Quantitative Research: Causal Effects, Time-Series Analysis and Large-Scale Data*

I. EDUCATION

National Taiwan University **September 2019 – August 2025**
M.A., B.A. in Economics

II. EXPERIENCE

Behavioral and Data Science Research Center (BDSRC) **July 2023 – July 2025**
Research Assistant

Project: Large-Scale Human Mobility Analysis

- Processed **1.6B** call detail records (**200 GB+**) in **2 hours** via parallelized PySpark parsing and GPU-accelerated cuDF joins.
- Cut user-defined-function runtime by up to **90%** (1 hr → 5 min) over **0.5M+** individuals with multi-core parallelism.
- **Deployed** rootless Docker on a shared multi-GPU server, isolating each user's containers.

Project: Stable Web Scraping Pipelines — Sports Records & Search-Trend Data

- Built a **50-year** panel of NFL and NBA player, team, and coach records via web scraping.
- **Collected** search-interest series for **100+** #MeToo cases via an unofficial Google Trends client.

Department of Economics, National Taiwan University **July 2023 – January 2024**
Research Assistant

Project: LLM-Automated Economic Policy Uncertainty Index

- **Classified** **7,000+** news articles by few-shot LLM prompting, and fine-tuned OpenAI models on **40,000+** labeled articles.
- **Collected** metadata for **0.5M+** articles via Playwright automation.

Project: StochasticFrontiers.jl refactor

- **Rebuilt** **7,000+** lines of Julia, decomposing **100+** large functions into concise units with multithreaded parallelization.

Institute of Economics, Academia Sinica **November 2022 – May 2024**
Research Assistant

Project: ML Treatment Assignment for Policy Evaluation

- **Assigned** **0.5M+** workers and **11M+** individuals to treatment and control groups via an XGBoost classifier.
- Ran **3,000+** hyperparameter-tuning experiments tracked in Optuna and MLflow.

III. PUBLICATIONS

Hsiu-Hsuan Yeh, Yu-Lieh Huang, Ziho Park, and Chung-Chi Chen. 2024. **Automation of Text-Based Economic Indicator Construction: A Pilot Exploration on Economic Policy Uncertainty Index**. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM '24)*. Boise, ID, USA.

IV. HONORS & AWARDS

Honorable Mention, Best Master's Thesis Award (Econometrics) — Taiwan Economic Association, Nov 2025.

V. SELECTED PROJECTS

Quantitative Finance

- [bitcoin-variance-analysis](#) — Decomposed Bitcoin price and volatility dynamics with Random Forest and GARCH, ranking drivers by Shapley value.
- [markov-switching](#) — Identified regime switches in cryptocurrency and equity series with Markov-switching models.

Machine Learning & NLP

- [fomc-hawkish-dovish](#) — Compared 3 pooling strategies ([CLS], mean, RNN) over frozen BERT representations for hawkish/dovish FOMC text classification, isolating the role of the NSP-pretrained pooling layer across BERT variants and fine-tuning corpora.
- [course-recommendation](#) — Course recommender ensembling 3 collaborative-filtering models (ALS, BPR, KNN) with BM25 retrieval and re-ranking; evaluated by MAP@50 for both seen and unseen users.
- [llm-research](#) — Minimal Python package over LangChain exposing a few-shot prompting API with MLflow tracking of LLM outputs.

Numerical & Scientific Computing

- [BasicIntegration.jl](#) — Julia integration package covering 4 schemes (Gaussian quadrature, Monte Carlo, quasi-Monte Carlo, importance sampling) for univariate and multivariate integrals, selecting the method automatically from the integration domain.
- [Jackcompiler-syntax-analyser](#) — Tokenizer and recursive-descent parser for the Jack language, emitting an XML syntax tree from raw source.

VI. SKILLS

Languages: Python, Julia

Machine Learning: PCA, XGBoost, Random Forest, Elastic Net, DBSCAN, BERT/PyTorch Lightning, Shapley values.

Econometrics: causal inference, policy evaluation, time-series analysis, GARCH, Markov-switching models.

Data Engineering: Polars, pandas, PySpark, RAPIDS cuDF, cuML, Joblib, Numba, MongoDB, httpx, Playwright.

Tools: Git, Docker (rootless, GPU), Linux (Arch), Neovim, LangChain, OpenAI fine-tuning.